2025-02-06-DSPonFPGALab1

title: EE3041 DSP on FPGA Big Project Lab 1 - Waveform Generator date: 2025-02-06 09:00:00 +0700 categories: [Lab Courses, EE3041 DSP on FPGA Big Project] tags: [lab, dsponfpga, fpga, courses] # TAG names should always be lowercase author: link toc: true comments: false description: Lab 1 for EE3041 DSP on FPGA Course at HCMUT media_subpath: /assets/img/DSPonFPGA/Lab1/

Lab 1 - Waveform Generator

Objective

The primary objective of this lab is to design and implement a versatile waveform generator using an FPGA. The generator should be capable of producing various types of waveforms with adjustable parameters such as frequency, amplitude, and duty cycle. Additionally, the system should be able to inject noise into these waveforms. The final implementation will be demonstrated on an FPGA using either the DE10 Standard Development Kit or the DE2 Standard Development Kit (students are free to choose), with the output displayed on an oscilloscope.

A Note about Collaboration

Big project is to be accomplished in groups. All work must be from your own team members.

Hints from others can be of great help, both to the hinter and the hintee.
Thus, discussions and hints about the assignment are encouraged. However, the project must be coded and written up in groups (you may not show, nor view, any source code from other students' groups). We **WILL** use automated tools to detect copying.

Use Emails/LMS to ask questions or come visit us (203B3) during office hours.
We want to see you succeed, but you have to ask for help.

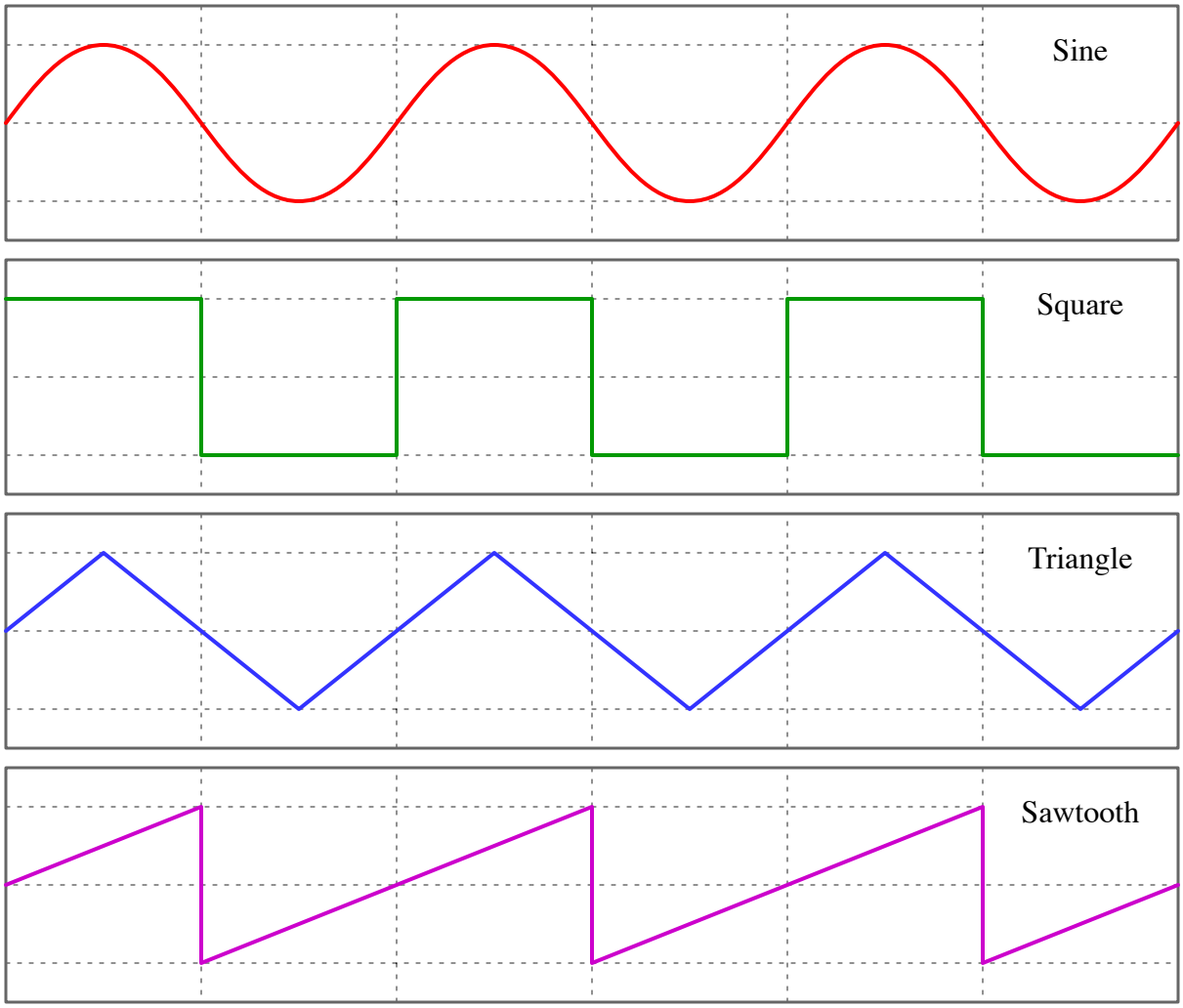
Overview

In this lab, you will design a waveform generator capable of producing a variety of waveforms such as sine, square, triangle, sawtooth, ECG, and noise. The generator should allow for the adjustment of key parameters including frequency, amplitude, and duty cycle (where applicable). Additionally, the system should be able to inject noise into the generated waveforms.

Background

Waveforms

Waveforms are graphical representations of the variation of a signal over time. Common types of waveforms include: - **Sine Wave**: A smooth, periodic oscillation. - **Square Wave**: A periodic waveform that alternates between two levels. - **Triangle Wave**: A periodic waveform that ramps up and down linearly. - **Sawtooth Wave**: A periodic waveform that ramps up linearly and then drops sharply. - **ECG Wave**: A waveform that mimics the electrical activity of the heart.

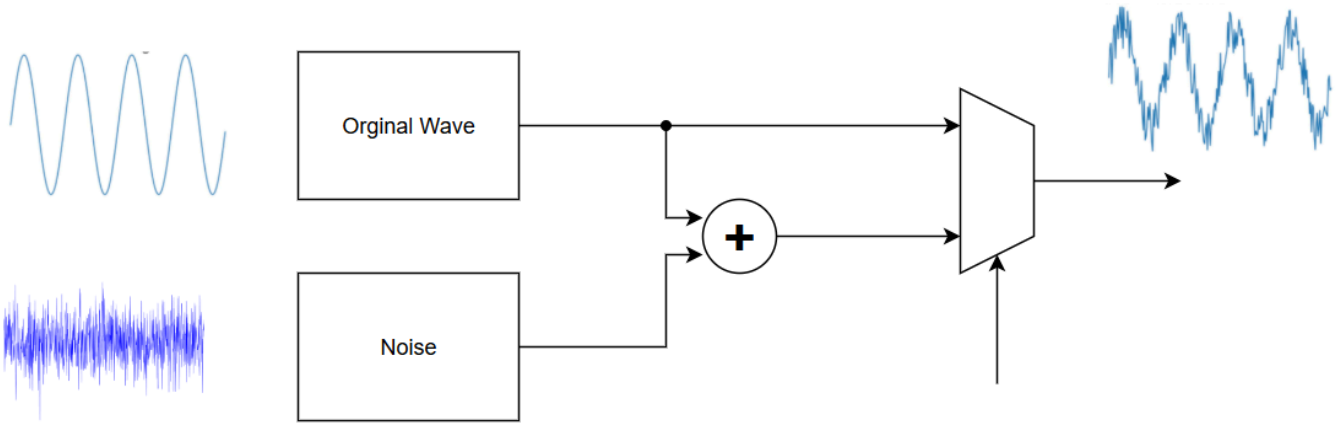


Waveform Generator

A waveform generator is an electronic device that can produce these various types of waveforms. In this lab, you will design a digital waveform generator using an FPGA, which allows for precise control over the waveform parameters.



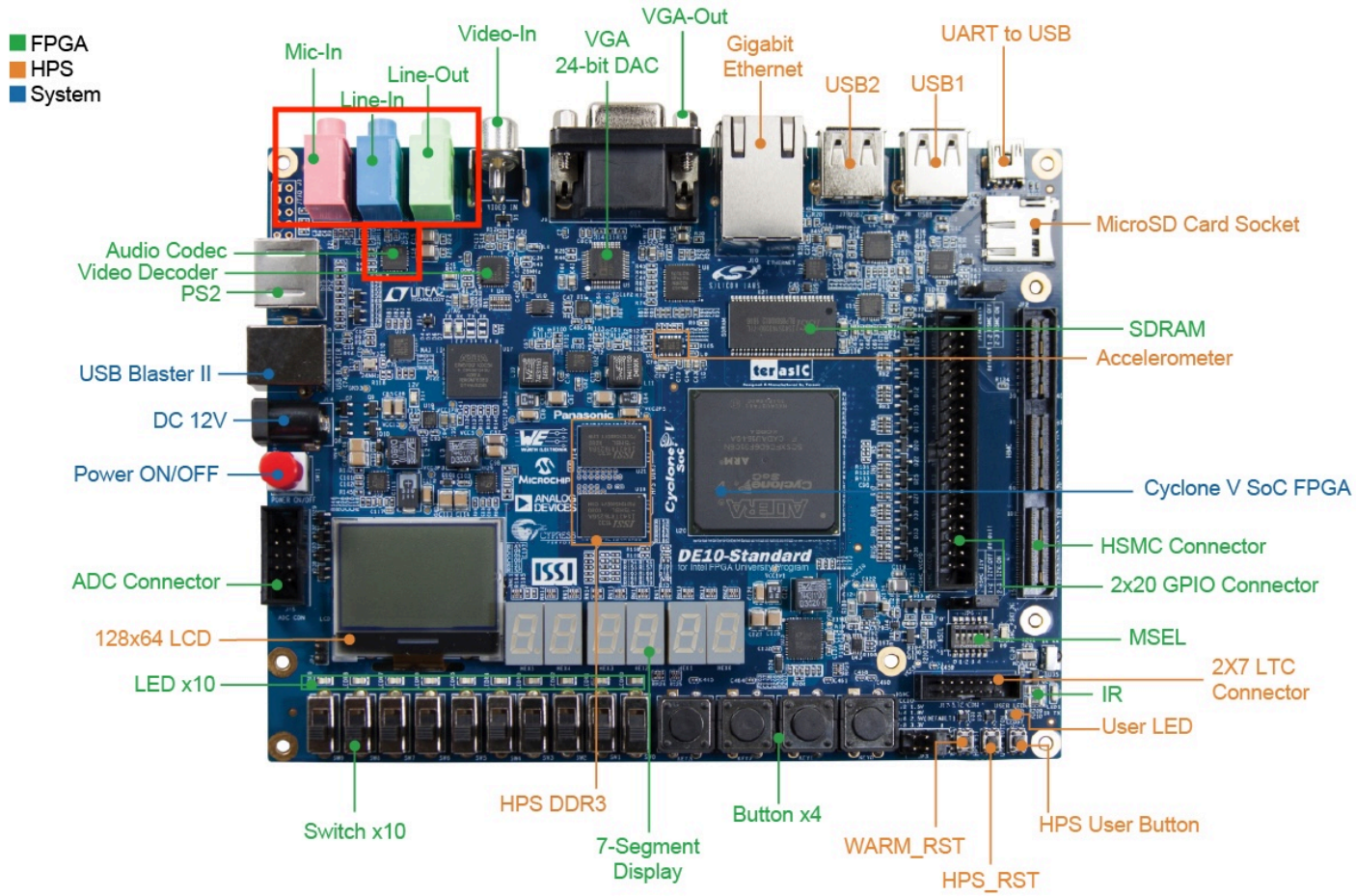
The waveform generator must include the capability to inject noise into the original waveforms. This feature is essential for Lab 2, where you will be required to filter out the injected noise using a designed filter. Ensure that the noise injection is controllable and can be adjusted in terms of frequency, amplitude, you can use a pseudo random generator (eg. LSFR) to generate the values.



DE10 Standard Development Kit

In this Lab 1, we will use the DE10 Standard Development Kit, a powerful FPGA development platform based on the Cyclone V SX SoC—5CSXFC6D6F31C6N. The DE10 kit offers a wide range of resources, including:

- 110K LEs, 41,509 ALMs
- 5,761 Kbits embedded memory
- 6 FPGA PLLs and 3 HPS PLLs
- 2 Hard Memory Controllers
- 112 DSP blocks
- 64MB (32Mx16) SDRAM on FPGA
- Wolfson WM8731 24-bit Audio CODEC



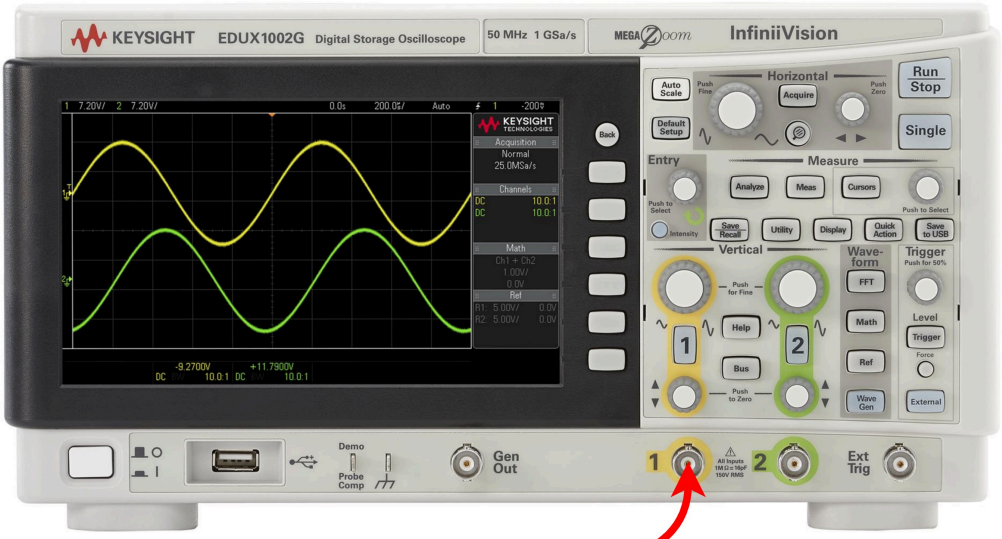
This makes it ideal for digital signal processing (DSP) tasks. In this lab, we will use the Audio CODEC as a DAC to output the 24-bit processed data through the audio output port. Additionally, in Lab 203B3, we provide a 3.5mm jack to BNC cable, which you can use to connect the output to an oscilloscope for visualizing the signal.

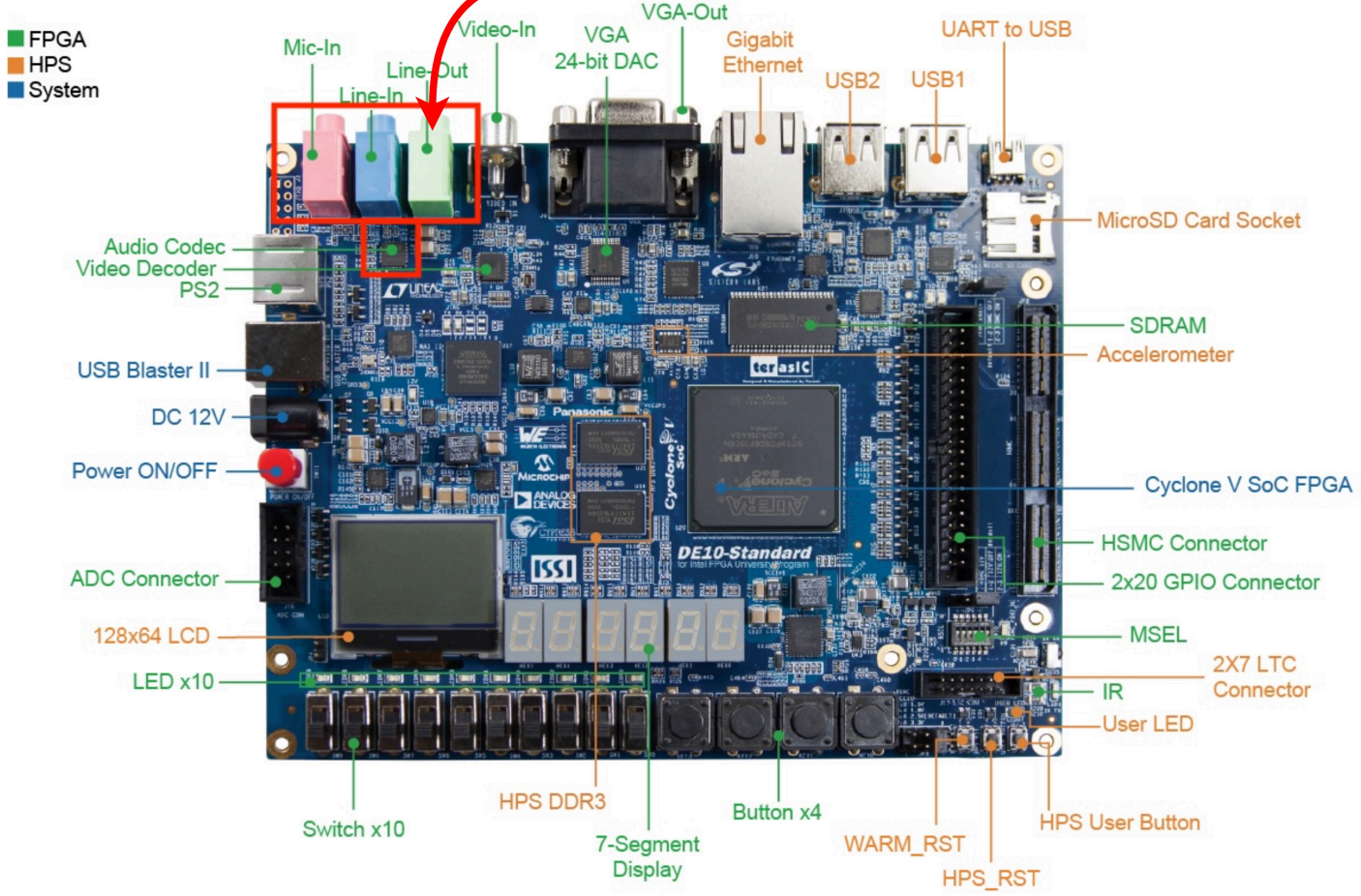


You can also buy your own cable: https://s.shopee.vn/7fQoF97hYY

Assignment Overview

1. **Design the Waveform Generator**:
    - Choose an appropriate hardware description language (HDL) for your design.
    - Create modules for generating different types of waveforms.
    - Implement controls for adjusting frequency, amplitude, and duty cycle.
    - Add functionality to inject noise into the waveforms.
2. **Simulate the Design**:
    - Use simulation tools to verify the correctness of your design.
    - Ensure that all waveforms and parameters can be correctly generated and adjusted.
3. **Implement on FPGA**:
    - Use either the DE10 Standard Development Kit or the DE2 Standard Development Kit to implement your design.
    - Connect the FPGA to an oscilloscope using the built-in CODEC and a 3.5-to-BNC cable.
    - Verify the output on the oscilloscope to ensure it matches the expected results.


KEYSIGHT EDUX1002G Digital Storage Oscilloscope
50 MHz 1 GSa/s
InfiniiVision
Horizontal
Run Stop
Single
Entry
Measure
Vertical
Waveform
Trigger
Demo
Probe Comp
Gen Out
Ext Trig


FPGA
HPS
System
Mic-In
Line-In
Line-Out
Video-In
VGA 24-bit DAC
VGA-Out
Gigabit Ethernet
USB2
USB1
UART to USB
MicroSD Card Socket
Audio Codec
Video Decoder
PS2
SDRAM
Accelerometer
USB Blaster II
DC 12V
Power ON/OFF
Cyclone V SoC FPGA
HSMC Connector
ADC Connector
2x20 GPIO Connector
128x64 LCD
MSEL
LED x10
2X7 LTC Connector
IR
User LED
Switch x10
HPS DDR3
7-Segment Display
Button x4
WARM_RST
HPS_RST
HPS User Button
DE10-Standard

Demonstration on FPGA

To showcase your waveform generator on either the DE10 Standard Development Kit or DE2 Standard Development Kit, your implementation must meet the following requirements:

- **Waveform Generation**: The system should generate multiple waveform types, including sine, square, triangle, sawtooth, and ECG. Users must be able to adjust frequency, amplitude, and duty cycle (if applicable) using switches and buttons. The implementation method is flexible.
- **Noise Injection**: The system should support injecting noise into the generated waveforms.
- **Waveform Visualization**: The output waveform must be displayed on an oscilloscope using the FPGA's built-in CODEC and a 3.5mm-to-BNC cable.

For Credit

Total: **10 points** (max), plus bonus.

Criterion (weight)	Emerging / Below Standard	Progressing / Approaching	Proficient / Meets Standard	Advanced / Exemplary
Specification & Design Understanding (1 pt)	No/incorrect spec; types or params missing	Spec partial; some types/params ambiguous	All waveform types (sine, square, triangle, sawtooth, ECG, noise) + params defined	Spec complete, justified, extensible
Baseline RTL Design (1 pt)	RTL incomplete or doesn't simulate	Compiles but partial sine/square/noise; 0 waveforms verified	Sine, square, noise injection implemented & verified for at least 1 waveform	RTL clean, robust, fully verified
Additional Waveforms (1.5 pts: 0.25 ea simple, 0.75 ea complex)	No extra waveforms beyond baseline	One simple waveform added & verified	Multiple simple waveforms and/or ECG added & verified	All extra waveforms (triangle, sawtooth, ECG) done, full credit
Simulation & Analysis (1.5 pts)	Sim missing or ignores input params	Sim runs; shallow accuracy/responsiveness analysis	Sim uses input params; explains accuracy & responsiveness	Thorough sim with plots, quantitative analysis, edge cases
Synthesis & Timing (2 pts)	Not synthesized / not synthesizable	Synthesizes but misses timing or report unanalyzed	Synthesizes, meets timing, report checked	Report deeply analyzed (utilization, fmax, slack), trade-offs discussed

Criterion (weight)	Emerging / Below Standard	Progressing / Approaching	Proficient / Meets Standard	Advanced / Exemplary
FPGA Live Demo (3 pts)	No demo or fails to show generation	Partial — some waveforms/params not working on HW	Live demo shows real-time generation + param control via oscilloscope	Demo polished, all types/params working, clear evidence
Bonus (uncapped)	No bonus attempted	—	Innovative improvement, solid suboptimal-result analysis, or social media content (e.g., short demo video/post) showcasing your project	Notable original enhancement + rigorous validated analysis + polished social media content with strong engagement

Reporting Requirements

Your report should be concise but detailed enough to clearly explain your design process, simulations, and results. Make sure to: - Show clear evidence of how your generator performs with the provided input signals, including graphs or waveforms and analysis. - Include the synthesis report and analyze the results, such as resource utilization, timing constraints, and operating frequency. - Provide an image of the oscilloscope output if you successfully implemented the design on the FPGA. - If the filter doesn't perform well, offer a detailed analysis of why the results are not as expected and propose improvements or solutions.

Your report should focus on quality rather than length. While it shouldn't be excessively long, it needs to be thorough and well-structured. Points will be deducted for poor reporting, such as missing key details, insufficient analysis, or unclear presentation. Make sure to present your work in a professional and readable format.

How To Turn In Your Solution

This semester we will be using LMS, simply submit the zip file with your reports and codes.

Demos and Late Penalty

We will have demo / presentation times on of class times on or near the due date. Since we will demo from the files in your zip, it is possible that you'll demo on a following day.

Define Late: Lateness is determined by the file dates of your submission on LMS.